

The Prior Is Not the Bottleneck: Where Success Comes From in Guided Diffusion Motion Planning

Anonymous
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Abstract

Guided diffusion is a popular paradigm for robot motion planning, and a steady stream of work improves the *generative prior*: better noise schedules, richer trajectory representations, and environment conditioning. We present a controlled study on Guided Polynomial Diffusion (GPD), re-implemented in the EDMP codebase, showing that for a guided planner **the prior is largely not the bottleneck**. Across five independent prior-side interventions—recalibrating the noise schedule, increasing polynomial capacity, adding a smoothness penalty, conditioning the denoiser on the scene with classifier-free guidance, and training $5\times$ longer (to 1M steps)—planning success rate (SR) never improves, with deltas ranging from statistically zero to small significant *decreases*, even when the interventions lower training loss or reconstruction error. A failure-mode analysis isolates *continuous (swept)* infeasibility as the dominant residual failure. Acting on it, two interventions on the *feasibility machinery*—faithful-collision RRT stitching (+14 *pp*) and a trajectory-optimization repair (+8.9 *pp*, 95% CI [7.6, 10.2])—break the plateau that no prior-side change could. A naive-seed control sharpens the claim: the prior is *necessary* as an initializer (a straight-line seed with the same repair reaches only 55% vs. 82%) but *improving* it does not help. The repair lever transfers to a second planner (EDMP, +10.3 *pp*) and collapses the diagnosed swept-collision failure (15.5% $\rightarrow$ 3.0%). We argue the community’s effort is mis-allocated and release an open, GPU-native re-implementation with all ablations.

1 Introduction

Diffusion models generate diverse, multimodal trajectories and, with classifier (cost-gradient) guidance, can be steered to avoid obstacles at inference time [10, 6, 12]. GPD [2] compresses a trajectory into eight Bernstein control points and shortens the diffusion chain to $T=64$ for fast planning. Most follow-up work—and the natural set of “obvious” improvements—targets the generative model: the noise/SNR schedule, the trajectory parameterization, and conditioning the denoiser on the environment.

We ran those improvements as controlled, single-variable experiments and found they **do not raise success rate** (Fig. 1). This paper reports that negative result, explains it, and then shows where the gains actually are: the inference-time feasibility machinery. Our central, somewhat contrarian claim is that for guided diffusion planners a competent prior is “good enough,” and the returns lie in collision-aware stitching and trajectory-optimization repair.

The stakes are practical. Retraining a prior—a new schedule, a larger basis, scene conditioning, or simply a longer run—costs GPU-days and is the default first move when a diffusion planner underperforms. If, as we find, that move is systematically ineffective while a retraining-free repair step yields double-digit gains, then the field’s effort is mis-directed. We make this case not with a single comparison but with five prior-side interventions and two feasibility levers, each measured under one continuous-execution success metric with confidence intervals, so the asymmetry in Fig. 1 cannot be attributed to noise or to a single favorable configuration.

Contributions.

1. A **decoupling result**: five prior-side improvements (including a $5\times/1\text{M}$ -step training budget) each improve a nominal generative metric yet never raise SR—each backed by a per-intervention confidence interval.
2. A **failure-mode diagnosis** isolating continuous/swept infeasibility as the dominant residual failure, and its *closure* after repair.
3. **Two positive levers on the feasibility machinery**: faithful-collision stitching (+14 *pp*) and trajectory-optimization repair (+8.9 *pp* on GPD, +10.3 *pp* on EDMP—planner-agnostic).
4. A **naive-seed control** scoping the claim: the prior is a necessary initializer, not a tunable lever.
5. An **open, GPU-native** re-implementation with every ablation.

2 Related Work

Diffusion for motion planning. Diffuser [10] framed trajectory generation as denoising; MPD [6] and EDMP [12] add cost-gradient (classifier) guidance to enforce collision avoidance at inference, EDMP using an ensemble of guidance costs. GPD [2] makes this fast by compressing trajectories to a low-degree Bernstein basis and shortening the diffusion chain. This line of work overwhelmingly improves the *prior*: the schedule, the representation, or scene conditioning. We instead hold the prior fixed and interrogate whether those improvements pay off.

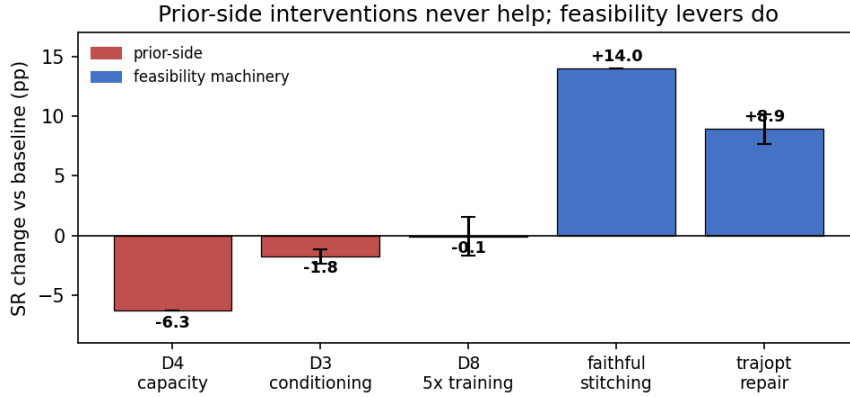


Figure 1: Success-rate change for each intervention on the hybrid benchmark. Every prior-side change (red) is zero or negative; both feasibility-machinery levers (blue) are large and positive. Error bars are 95% confidence intervals where multi-seed runs exist. This asymmetry—prior improvements never help, feasibility repair does—is the paper’s central finding.

Feasibility repair. A parallel thread refines a learned or sampled seed with trajectory optimization: PRESTO [4] and MPD [6] warm-start an optimizer from a diffusion sample. We adopt this mechanism but use it as an *analytical instrument*—contrasting it against five prior-side changes under one metric—rather than as a standalone method.

Learned and classical baselines. MPiNets [7] regresses policies from large demonstration sets, and cuRobo [13] is a GPU-parallel classical planner; both define the strong end of the speed/quality frontier we position against qualitatively (our absolute SR is below published systems; §3).

3 Background and Setup

GPD. A TemporalUNet ϵ -predictor (~ 3.95 M params) over eight Bernstein control points, $T=64$, variance schedule $\beta = \text{linspace}(0, 0.02, T+1)$; start/goal are pinned at the boundary control points. Inference samples K candidates under cost-gradient guidance (differentiable AABB intersection-volume plus swept-volume via PyTorch forward kinematics), then selects and stitches them [2, 8].

Success metric. We execute the trajectory in PyBullet under position control; success requires *no obstacle contact during execution*. This is a continuous-execution test, not a waypoint check.

Setup. Unless noted, numbers are on the MPiNets [7] *hybrid*-solvable set (Fig. 2), balanced 100 scenes per environment type (400 total), under the GPD-stitched configuration (single guide, $K=32$, faithful-collision RRT-Connect stitching [11]). Our re-implementation reproduces the method ordering of [2] but sits ~ 21 pp below its absolute numbers; all claims below are therefore *deltas measured within our own system*, never cross-method absolutes. The gap is not under-training (App. A.4): the GPD paper releases no code or weights and leaves K , guidance scale, and cost formulas unspecified, and in-system tuning recovers only ~ 3 of the ~ 20 pp.

4 The Prior Is Not the Bottleneck

We test the five most natural prior-side improvements. Each is a clean single-variable change sharing an identical guidance/stitching pipeline.

4.1 Noise-schedule recalibration

At $T=64$ the linear schedule gives $\bar{\alpha}_T \approx 0.52$ —the model is never trained near pure noise, while inference starts from $\mathcal{N}(0, I)$. The “fix” (raise terminal noise, or use cosine) *lowered* SR on all scene types despite a *lower* training loss ($0.53 \rightarrow 0.43$). The low-terminal-noise regime keeps the reverse process near-identity, acting as an implicit smoothness prior; injecting variance yields less-smooth, more-colliding paths. ϵ -loss is a poor proxy for planning SR.

4.2 Polynomial capacity

Bernstein reconstruction error shows degree-7 ($n=8$) discards $5\text{--}17^\circ$ of joint motion on the constrained demonstrations (mean 5.47°), so the representation *is* lossy where GPD struggles. Yet training $n=16$ ($4\times$ lower reconstruction error, matched convergence loss) yields a **net** -6.3 pp: it beats $n=8$ on sharp-turn dresser scenes but loses on tabletop/cubby, and costs $5\times$ the training.

Mechanism (compression = smoothness regularizer). Measuring the raw sampled trajectories (100 scenes, before any repair), the compact $n=8$ prior is markedly smoother: joint-jerk RMS 0.00166 vs. 0.00214 (-22% ; Fig. 3), with a larger clearance margin. The low-degree basis cannot express the high-frequency motion a higher-capacity model fits, so it acts as an implicit smoothness prior—and smoother trajectories collide less under continuous execution. Reconstruction fidelity is *anti-correlated* with the smoothness that drives SR.

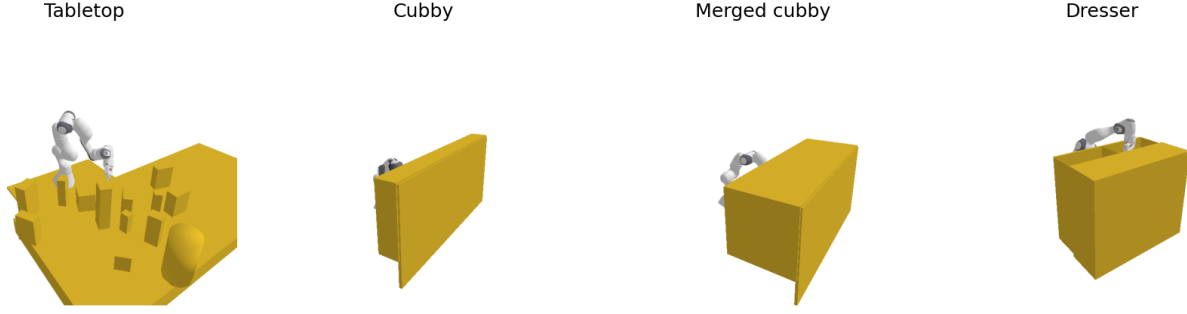


Figure 2: The four MPiNets [7] environment types, with the Franka Panda at a goal configuration reaching into each scene (obstacles in yellow). Tabletop scenes scatter small objects; cubby and merged-cubby require reaching through shelf openings; dresser requires reaching into a box—the constrained types where swept infeasibility dominates.

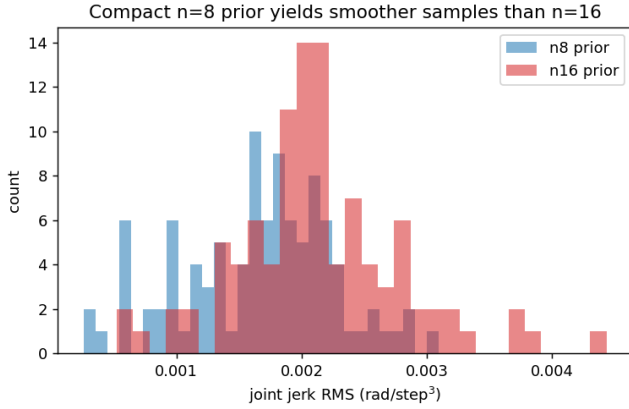


Figure 3: Distribution of raw-sample joint jerk over 100 hybrid scenes. The compact $n=8$ prior is smoother than the higher-capacity $n=16$ prior, a direct mechanism for why capacity hurts SR despite lower reconstruction error.

4.3 Scene-conditioned denoiser (CFG)

We add a masked-DeepSets [15] obstacle encoder whose embedding feeds the time embedding, with a learned null embedding for classifier-free guidance [9]. The conditioned model reaches a *lower* loss ($0.502 < 0.52$), yet with guidance removed the conditioned prior plans no better than the scene-blind prior: scene information enters through *guidance*, not the prior. Over five paired seeds the conditioning delta is -1.75 pp (95% CI $[-2.37, -1.13]$)—a small but significant *decrease*.

4.4 Training to convergence ($5\times$, 1M steps)

The most basic prior-side lever is more training. We train the canonical $n=8$ prior for 1,000,000 steps ($5\times$ the baseline); ϵ -loss drops $\sim 4\%$ and flattens for the final 300k steps. Over five paired seeds the delta is -0.10 pp (95% CI $[-1.71, +1.51]$)—statistically zero. This rules out under-training as the explanation for both the flat prior-side results and the reproduction gap.

4.5 Summary

Five prior improvements, five nominal-metric gains, **no positive SR gain**: deltas span statistically zero ($5\times$ training) to small significant decreases (conditioning -1.75 pp; schedule and capacity larger). Against a measured baseline sampling noise $\sigma \approx 1.6$ pp, no prior intervention clears it upward, while the repair lever below clears it by $\sim 6\sigma$.

5 Where Success Actually Comes From

5.1 Failure-mode analysis, and its closure

We categorize 200 hybrid failures of the baseline, and re-categorize after repair:

failure mode	baseline %	after repair %
success	74.0	78.5
goal/start IK infeasible	2.5	1.5
path_collision (waypoints)	8.0	17.0
dynamic_only (swept)	15.5	3.0

The executor and static checker share a contact model, so `dynamic_only` is genuine: the robot *sweeps through obstacles between waypoints* (Fig. 4). It is the dominant residual failure, and the repair we build (§5.3) collapses it to 3.0%. Of the eliminated swept failures, ~ 4.5 pp convert to success and the rest become statically-detectable `path_collision`: on the hardest scenes the optimizer churns but leaves a residual static collision—it converts dynamic failures, it does not manufacture them.

5.2 Lever 1: faithful-collision stitching

The stitcher originally certified bridges with the loose AABB intersection-volume *proxy*, marking colliding bridges as free. Replacing it with a faithful PyBullet check inside RRT-Connect yields $+14$ pp over the proxy plateau, the first intervention to move SR.

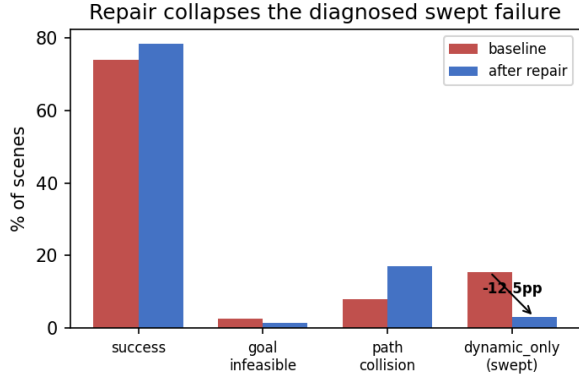


Figure 4: Failure-mode distribution before and after repair. The dominant baseline failure—*dynamic_only* swept collision—collapses from 15.5% to 3.0%, the very failure the repair objective (Eq. 1) targets.

5.3 Lever 2: trajectory-optimization repair

Motivated by the analysis above and recent work [4, 6], we add a post-hoc optimizer. With endpoints q_0, q_N pinned to start/goal, we solve over the interior waypoints $q_{1:N-1}$

$$\min_{q_{1:N-1}} \sum_i c(q_i) + \lambda_m \sum_i c\left(\frac{q_i + q_{i+1}}{2}\right) + \lambda_s \sum_i \|q_{i-1} - 2q_i + q_{i+1}\|^2, \quad (1)$$

where $c(\cdot)$ is the guide’s differentiable collision cost. The first term drives waypoints out of collision; the second penalizes collision at *edge-midpoints*, directly targeting the swept infeasibility diagnosed above; the third is a smoothness (acceleration) penalty. We run Adam with joint-limit projection and *early-stop* on a densified faithful collision check, returning a densified trajectory for overshoot-free execution. It reuses the existing differentiable cost—no retraining.

Over five seeds, repair raises SR from 72.9 ± 1.6 to 81.8 ± 1.3 : a paired $+8.9pp$, **95% CI** [7.6, 10.2], $t(4)=19.4$, $p<0.0001$. The optimizer targets a densified static check; the benchmark independently verifies dynamic execution, so this is a real executable-SR gain at $\sim +1.8s$ /scene (only on scenes needing repair).

5.4 Is the prior even needed? A naive-seed control

If improving the prior never helps, is the prior irrelevant? We replace the diffusion candidates with a single straight-line joint-space seed and run the *identical* downstream (RRT stitch $\pm$ repair):

seed	stitch only	stitch + repair
linear (no prior)	18.5 ± 0.3	55.0 ± 1.0
diffusion (prior)	72.9 ± 1.6	81.8 ± 1.3

The prior is **necessary as an initializer**: even the best repair on a linear seed reaches only 55%, $-26.8pp$ below

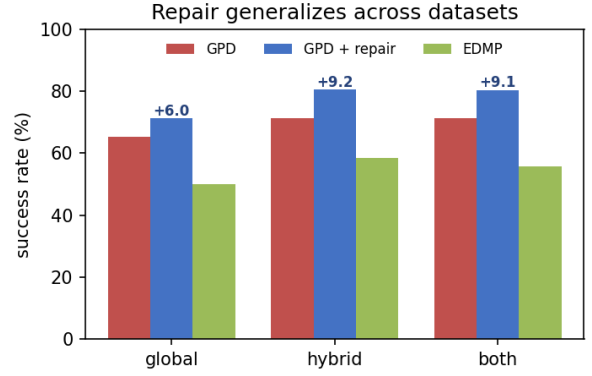


Figure 5: Repair generalizes across all three MPiNets solvable splits (+6 to +9 *pp*), and GPD+repair beats EDMP everywhere at a fraction of its planning time (Table below).

diffusion+repair—repair cannot substitute for a competent seed. Yet repair does far more *work* on the bad seed (+36.5 *pp*) than on the diffusion seed (+8.9 *pp*): good initialization and good repair are complementary, not interchangeable. This scopes the paper’s claim precisely—*improving* an already-competent prior buys nothing, but the prior is not dispensable.

5.5 The repair lever generalizes (EDMP)

If the bottleneck is the feasibility machinery, the same repair should help a different planner. Applied unchanged to EDMP [12] (a different model: $T=255$ over full 50-waypoint trajectories), repair lifts SR from 57.2% to 67.5% (+10.3 *pp*; cubby +18, merged +12, dresser +10)—*larger* than the GPD gain and on the same constrained scene types. The lever is planner-agnostic.

5.6 Generalization across datasets

The repair lever is not a hybrid-only artifact (Fig. 5). On the three MPiNets solvable splits (600 natural-weighted scenes each), it lifts SR by +6 to +9 *pp* everywhere, and consistently beats EDMP at a fraction of its planning time:

split	GPD	GPD+repair	EDMP
global	65.2	71.2 (+6.0)	50.0
hybrid	71.3	80.5 (+9.2)	58.5
both	71.2	80.3 (+9.1)	55.7
avg time	7.6s	9.4s	47.6s

The gain is largest on the constrained (hybrid/both) splits where swept infeasibility dominates, consistent with the failure-mode diagnosis.

6 Discussion

The two effective levers are both on the *inference-time feasibility machinery*, and both concern *continuous* feasibility

(collision-free bridges; collision-free swept motion). The ineffective levers are all on the prior, and several improved nominal generative metrics. A recurring theme is that *fidelity is decoupled from success at every level*: ϵ -loss (priors), reconstruction error (capacity), and even the differentiable repair *objective* (an exact sphere-SDF objective ties the cheap AABB proxy; App. A.2), because the faithful check is the true oracle. For guided diffusion planners, effort spent improving priors yields little once a competent guided prior exists; the returns are in feasibility repair.

When would the prior matter? Our claim is scoped, not absolute. The naive-seed control (§5.4) shows the prior is indispensable as an initializer—remove it and repair stalls at 55%—so a prior good enough to place the seed in roughly the right homotopy class is a prerequisite, not a target for further tuning. We expect prior improvements to pay off in regimes our benchmark does not stress: much higher-DoF systems, tasks with hard end-effector constraints the guidance cost cannot express, or settings without a fast faithful collision checker to serve as the repair oracle. Absent those, the empirical picture is stark—five prior-side changes, no gain; two feasibility changes, double-digit gains.

Limitations. Core results are on hybrid-solvable, balanced 400; we generalize the repair lever to global/both splits and to EDMP, but larger N and more planners would tighten the claim. Absolute SR is below the published GPD system; we therefore report deltas within a fixed system rather than cross-method absolutes.

7 Conclusion

In a controlled study of guided polynomial diffusion planning, the generative prior is not the bottleneck: five natural prior improvements never raise SR while improving nominal metrics, and a $5\times$ prior changes nothing. The gains live in the feasibility machinery—faithful-collision stitching (+14 *pp*) and trajectory-optimization repair (+8.9 *pp* on GPD, +10.3 *pp* on EDMP)—which also collapses the diagnosed swept-collision failure. The prior earns its place as a necessary initializer, not as a tunable lever. We release the implementation and all ablations to support feasibility-centric research on diffusion planners.

A Ablations

A.1 Trajopt hyperparameters (time/accuracy Pareto)

One-factor-at-a-time around the default (iters 60, densify 3, midpoint-weight 1.0), hybrid 400 (Fig. 6). *Iterations peak, not plateau*: SR saturates by ~ 40 and iters 100 regresses (-3.2 *pp*)—over-optimization drifts waypoints off the smooth seed. The *midpoint* term is causal for swept feasibility (-3.4 *pp* without it). Output densification peaks at 3.

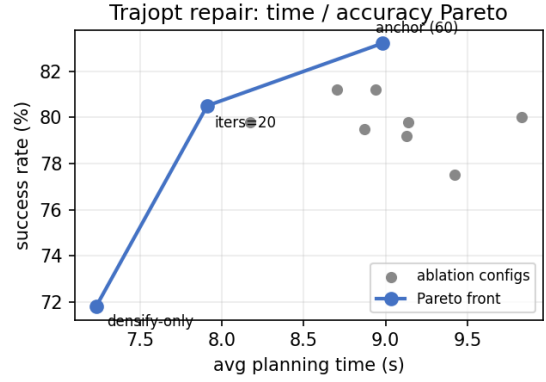


Figure 6: Time/accuracy trade-off of the trajopt-repair configurations. The Pareto front runs densify-only $\rightarrow$ iters 20 $\rightarrow$ the anchor (iters 60); all other settings are dominated.

knob	value	SR%	avg t (s)
iters	0 (densify-only)	71.8	7.2
	20	80.5	7.9
	40	81.2	8.7
	60	83.2	9.0
	100	80.0	9.8
out-densify	1	77.5	9.4
	3	83.2	9.0
	5	79.5	8.9
mid-weight	0.0	79.8	8.2
	1.0	83.2	9.0
	2.0	81.2	8.9

A.2 Exact vs. proxy collision objective

We implemented an exact GPU-batched sphere-SDF objective (FK-placed link spheres vs. oriented obstacle boxes), removing the AABB double-inflation. Swapped into the same repair loop (3 seeds) it gives 82.0 ± 0.9 vs. 81.8 ± 1.3 for the proxy—a paired $+0.6$ *pp*, not significant, at equal time. Because the faithful PyBullet check is the stopping oracle, objective fidelity is not a lever.

A.3 Per-intervention seed CIs

Five paired seeds, hybrid 400. Baseline 72.9 ± 1.6 ; conditioning 71.2 ± 1.3 ($\Delta = -1.75$, CI $[-2.37, -1.13]$); $5\times$ training 72.8 ± 0.9 ($\Delta = -0.10$, CI $[-1.71, +1.51]$); repair 81.8 ± 1.3 ($\Delta = +8.9$, CI $[7.6, 10.2]$).

A.4 Reproduction study: how far in-system tuning closes the gap

Our re-implementation reproduces the GPD paper’s method ordering but sits ~ 21 *pp* below its absolute numbers (§3). Because all our contributions are within-system deltas, absolute matching is not required; this study asks how much of the gap

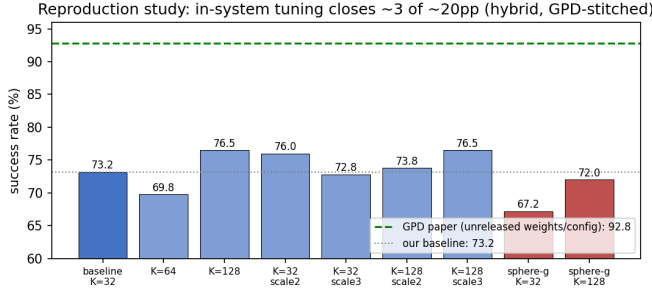


Figure 7: Reproduction study on the GPD-stitched config (hybrid). In-system candidate and guidance-scale tuning recovers only ~ 3 of the ~ 20 pp gap to the GPD paper (green, unreleased weights/config); exact sphere-SDF guidance (red) is worse.

is recoverable in-system and attributes the residual. Crucially, the GPD paper releases **no code and no weights**, and leaves the guidance scale, candidate count K , and collision-cost formulas unspecified—so an exact match is not reconstructable from the paper. Training is not the cause: our budget already matches theirs (20k steps, same 6.54M Bernstein set), and $5\times/1$ M-step training moves SR by -0.1 pp (§A.3).

Sweeping the two documented levers on the GPD-stitched config (hybrid, 400 scenes; Fig. 7):

config	SR%	avg t (s)
baseline (K=32, scale 1)	73.2	7.3
K=64	69.8	8.4
K=128	76.5	10.9
K=32, scale 2	76.0	7.3
K=32, scale 3	72.8	7.3
K=128, scale 3	76.5	10.9
K=32 + sphere-SDF guidance	67.2	3.4
K=128 + sphere-SDF guidance	72.0	4.1
GPD paper (unreleased weights/config)	92.8	1.9

Raising the candidate budget to $K=128$ and the guidance scale to $2\times$ each add ~ 3 pp, plateauing at **76.5%** — recovering only ~ 3 of the ~ 20 pp gap. Replacing the guide’s inflated-AABB gradient with our exact sphere-SDF gradient (App. A.2) as the *denoising guidance* is worse (-6 pp): the guide’s clearance/expansion schedule supplies a safety margin that geometric exactness does not, a further instance of objective-fidelity being decoupled from SR. The residual ~ 16 pp is thus attributable to factors we cannot reconstruct from the paper — the tuned 7-cost guidance ensemble (whose $1G \rightarrow 7G$ gain we reproduce at $+6$ pp vs. the paper’s $+18$ pp), the exact guidance scale, and the authors’ trained weights. This confirms that absolute cross-method comparison is unavailable here and that our within-system deltas are the sound unit of claim.

A.5 Continuous-feasibility audit and a dense repair objective

The dominant failure (§3, App. A.3) is *swept* infeasibility, yet the executor and our checks only sample the path discretely. Dif-

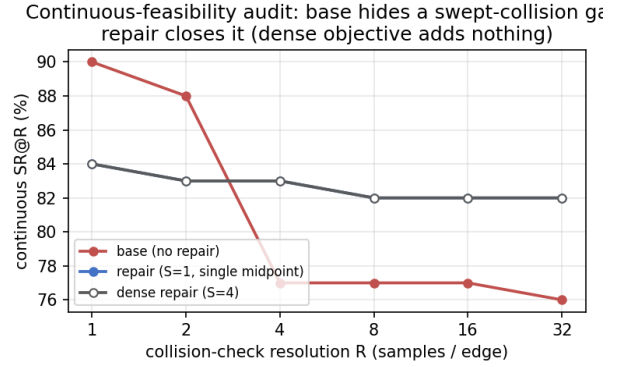


Figure 8: Continuous-feasibility audit (100 hybrid scenes). The base planner’s success rate collapses as the collision check densifies (90% at waypoints $\rightarrow$ 76% at $R=32$): ~ 14 pp of “waypoint-free” plans actually sweep through obstacles. Repair nearly eliminates this ($84 \rightarrow 82\%$, flat), and a dense $S=4$ objective coincides exactly with the single-midpoint $S=1$ repair.

ferentiable swept-volume models exist [1, 3] but none are used inside a diffusion planner’s repair loop; likewise safe-diffusion methods [14] enforce constraints only at sampled points. We therefore (i) *audit* continuous feasibility at increasing check resolution R (samples per edge), and (ii) test a *dense* repair objective that evaluates collision at S interpolated sub-configs per edge (the §5.3 midpoint is $S=1$).

The audit exposes a real swept gap, which repair closes. Fig. 8: the *base* planner is 90% feasible at waypoints but only 76% under a dense ($R=32$) continuous check — roughly a seventh of nominally-free plans sweep through obstacles between waypoints. Trajectory-repair is nearly *resolution-flat* ($84 \rightarrow 82\%$): it produces genuinely continuous-feasible paths, not just waypoint-free ones.

But a denser repair objective does not help. Over 5 seeds (hybrid 400): $S=1$ gives 81.8 ± 1.3 , $S=4$ gives 81.7 ± 1.3 — a paired -0.15 pp (95% CI $[-0.79, +0.49]$) — while costing $+2.5$ s/scene ($9.0 \rightarrow 11.4$ s; $S=8$ is 14.8s). The single-midpoint swept term is already a sufficient statistic for continuous feasibility: because repair early-stops on a *densified faithful* oracle, a denser *differentiable* objective changes nothing — the same objective-fidelity $\perp$ SR decoupling seen for the noise schedule, capacity, and the exact sphere-SDF objective (App. A.2). Continuous feasibility is worth *measuring* and *repairing*, but not worth a heavier objective.

The audit generalizes across datasets and planners. Fig. 9: the waypoint $\rightarrow$ dense success gap is universal — base GPD loses 27/18/21 pp on global/hybrid/both, and EDMP loses 10 pp on hybrid — and in every case repair flattens the curve and dynamic execution tracks the dense static check. This makes the audit a general evaluation-rigor prescription: learned planners should report success at a stated continuous-check resolution (classical planners already do [11]), not at the waypoints alone.

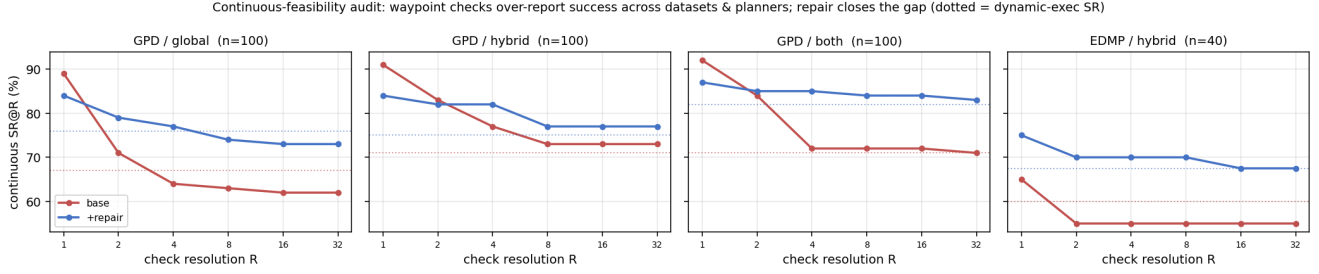


Figure 9: Continuous-feasibility audit across datasets (GPD: global/hybrid/both) and a second planner (EDMP). Base (red) success collapses as the check densities; repair (blue) is nearly resolution-flat; dotted lines are dynamic-execution SR ($\approx$ the dense static check). The waypoint $\rightarrow$ dense over-reporting is 10–27 *pp*.

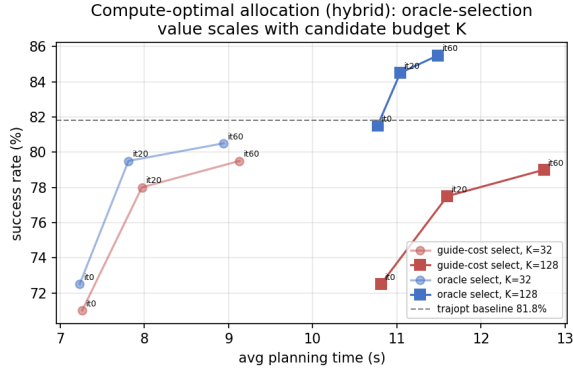


Figure 10: Compute-optimal allocation (hybrid). Oracle (verifier) selection’s benefit grows with candidate budget K , but the 5-seed confirmation of the best point ($K=128$, +trajopt) is 83.5 ± 1.0 — statistically tied with the trajopt baseline (dashed) at higher cost.

A.6 Can the feasibility lever be shortcut? Two more negative levers

Beyond a denser objective, we tested two literature-motivated ways to improve/replace the stitch+trajopt machinery. Both fail to beat it, reinforcing that it is the lever.

Verifier-guided selection (best-of- K by the faithful oracle). Since the faithful PyBullet oracle—not the differentiable cost—gates success, we select the candidate with the highest densified free-fraction (a best-of- N verifier [5]) rather than the guide’s cost. At $K=32$ this *underperforms* guide-cost+RRT-stitch (-3.4 *pp* with repair, 5 seeds), because stitching *bridges* colliding segments across candidates while selection keeps only one. Its value scales with K (Fig. 10): at $K=128$ oracle-selection+trajopt reaches 83.5 ± 1.0 vs. the 81.8 ± 1.3 trajopt baseline — a paired $+1.65$ *pp* (95% CI $[-1.0, +4.3]$, *not significant*) at $+3$ s/scene. A compute-optimal-allocation grid over K , guidance, and repair iterations (per-component wall-clock: diffusion dominates at ~ 7 – 10 s, repair ~ 1 – 3 s, selection < 0.5 s) shows the Pareto front prefers oracle-selection but the frontier does not clear the trajopt baseline by a significant margin.

Learned amortized repair. We distilled the iterative trajopt into a one-shot network (stitched seed + obstacle set $\rightarrow$ residual repair; DeepSets scene encoder, endpoints pinned) from 1800

(seed, repaired) pairs, to reclaim latency [4]. It *fails*: held-out regression loss exceeds the identity (copy-the-seed) baseline, and in the pipeline one-shot learned repair scores 64.2 ± 2.1 — *worse than no repair* (-8.7 *pp*) — while a learned+trajopt warm-start hybrid reaches only 75.6 ± 1.8 ($<$ the 81.8 trajopt baseline) at barely lower latency. Distillation from this data budget does not generalize; the iterative oracle-gated optimization remains necessary.

Synthesis. Across four attempts to improve or shortcut the feasibility machinery — denser continuous objective, exact sphere-SDF objective/guidance, verifier-guided selection, and learned amortized repair — none significantly exceeds plain oracle-gated trajopt repair. The lever is robust; the returns are in *using* the faithful oracle inside stitch/repair, not in a heavier objective, a cleverer selector, or a distilled shortcut.

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